Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The set of values of x for which $\sin x \cdot \cos^3 x > \cos x \cdot \sin^3 x$ in $[0, 2\pi]$, is

(A) $(0, \pi)$

(B) $\left(0, \frac{\pi}{4}\right)$

(C) $\left(\frac{\pi}{4}, \pi\right)$

(D) none of these

Q.2

If $\sin\theta = a$ for exactly one value of $\theta \in \left[0, \frac{7\pi}{3}\right]$, then a can be

(A) $\frac{\sqrt{3}}{2}$

(B) $\frac{1}{2}$

(C) 0

(D) -1

Q.3

The values of $x \in \left(0, \frac{\pi}{2}\right)$ satisfying the equation $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ are

(A) $\frac{\pi}{36}, \frac{\pi}{12}$

(B) $\frac{\pi}{12}, \frac{11\pi}{36}$

(C) $\frac{12\pi}{36}, \frac{11\pi}{36}$

(D) $\frac{11\pi}{36}, \frac{\pi}{36}$

Q.4

The number of solutions of the equation $\tan x \cdot \tan 4x = 1$ for $0 < x < \pi$ is

(A) 2

(B) 4

(C) 6

(D) none of these

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Q.5

The most general solution of $\tan\theta = -1$ and $\cos\theta = \frac{1}{\sqrt{2}}$ is

(A) $n\pi + \frac{7\pi}{4}$

(B) $n\pi + (-1)^n \frac{7\pi}{4}$

(C) $2n\pi + \frac{7\pi}{4}$

(D) none of these

Q.6

The number of solution of equation, $e^{\cos^2 x} - e^{-\cos^2 x} + 4 = 0$ in $[0, 2\pi]$ is/are

(A) 1

(B) 2

(C) 3

(D) none of these

Q.7

If $2\sec 2\alpha = \tan\beta + \cot\beta$, then one of the values of $\alpha + \beta$ is

(A) π

(B) $n\pi - \frac{\pi}{4}, n \in Z$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{2}$

Q.8

The general solution of $\sin^2\theta\sec\theta + \sqrt{3}\tan\theta = 0$ is

(A) $\theta = n\pi + (-1)^{n+1}\frac{\pi}{3}, \theta = n\pi; n \in I$

(B) $\theta = n\pi; n \in I$

(C) $\theta = \frac{n\pi}{2}, n \in I$

(D) $\theta = n\pi + (-1)^{n+1}\frac{\pi}{3}, n \in I$

Q.9

The solutions of the equation $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$ are given by

(A) $x = n\pi \pm \frac{\pi}{2}; x = 2m\pi \pm \frac{\pi}{4}; x = k\pi \pm \frac{\pi}{6}$; where $n, m, k \in Z$

(B) $x = 2n\pi \pm \frac{\pi}{2}; x = m\pi \pm \frac{\pi}{4}; x = 2k\pi \pm \frac{\pi}{6}$; where $n, m, k \in Z$

(C) $x = n\pi \pm \frac{\pi}{2}; x = m\pi \pm \frac{\pi}{4}; x = k\pi \pm \frac{\pi}{6}$; where $n, m, k \in Z$

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(D) $x=(2n-1)\pi \pm \frac{\pi}{2}; x=m\pi \pm \frac{\pi}{4}; x=k\pi \pm \frac{\pi}{6}$; where $n, m, k \in Z$

Q.10

Let $\theta \in (\pi/4, \pi/2)$, which of the following statements is true?

(A) $(\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta} < (\cos\theta)^{\sin\theta}$

(B) $(\cos\theta)^{\sin\theta} < (\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta}$

(C) $(\sin\theta)^{\cos\theta} < (\cos\theta)^{\cos\theta} < (\cos\theta)^{\sin\theta}$

(D) $(\cos\theta)^{\cos\theta} < (\cos\theta)^{\sin\theta} < (\sin\theta)^{\cos\theta}$

Q.11

The number of solutions of the equation $\sin 5x \cos 3x = \sin 6x \cos 2x$ in the interval $[0, \pi]$ are

(A) 3

(B) 4

(C) 5

(D) 6

Q.12

The general solution of $\tan 3x = 1$ is

(A) $n\pi + \frac{\pi}{4}$

(B) $\frac{n\pi}{3} + \frac{\pi}{12}$

(C) $n\pi$

(D) $n\pi \pm \frac{\pi}{4}$

Q.13

Let $A = \sin^8\theta + \cos^{14}\theta$, then for all real θ

(A) $A \geq 1$

(B) $0 < A \leq 1$

(C) $\frac{1}{2} \leq A \leq \frac{3}{2}$

(D) none of these

Q.14

The complete solution of the equation $7\cos^2 x + \sin x \cos x - 3 = 0$ is given by

(A) $n\pi + \frac{\pi}{2} (n \in I)$

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(B) $n\pi + \frac{\pi}{4} (n \in I)$

(C) $n\pi + \tan^{-1}\left(\frac{4}{3}\right) (n \in I)$

(D) $n\pi + \frac{3\pi}{4}, k\pi + \tan^{-1}\left(\frac{4}{3}\right) (n, k \in I)$

Q.15

For all $x, y \in R$, the maximum value of $\sin^2 x + \sin^2 y + \sin^2(x + y)$ equals

(A) $\frac{3}{4}$

(B) $\frac{5}{4}$

(C) $\frac{7}{4}$

(D) $\frac{9}{4}$

Q.16

If 1 lies between the roots of the quadratic equation $3x^2 - 3\sin\theta x - 2\cos^2\theta = 0$. Then

(A) $\frac{\pi}{3} < \theta < \frac{5\pi}{6}$

(B) $n\pi < \theta < 2n\pi$

(C) $2n\pi + \frac{\pi}{6} < \theta < 2n\pi + \frac{5\pi}{6}$

(D) none of these

Q.17

The equation $(\cos a - 1)x^2 + (\cos a)x + \sin a = 0$ in x has real roots, then possible set of values of x may be

(A) $[0, 2\pi]$

(B) $[-\pi, 0]$

(C) $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

(D) $[0, \pi]$

Q.18

If $1 + \sin\theta + \sin^2\theta + \ldots\ldots$ to $\infty = 4 + 2\sqrt{3}, 0 < \theta < \pi, \theta \neq \frac{\pi}{2}$, then θ is equal to

(A) $\frac{\pi}{6}$ only

(B) $\frac{\pi}{3}$ only

(C) $\frac{\pi}{3}$ or $\frac{\pi}{6}$

(D) $\frac{\pi}{3}$ or $\frac{2\pi}{3}$

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Q.19

If $\tan x < -\sqrt{3}$ and $\sin x > \frac{\sqrt{3}}{2}$, then which of the following is true

(A) $x \in \left(\frac{\pi}{3}, \frac{2\pi}{3}\right)$

(B) $x \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

(C) $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$

(D) none of these

Q.20

The values of x satisfying the system of equations $2^{\sin x+\cos y} = 1, 16^{\sin^2 x+\cos^2 y} = 4$ are given by
(i) $x = n\pi + (-1)^n \frac{\pi}{6}$ and $y = 2n\pi \pm \frac{\pi}{3}$
(ii) $x = n\pi + (-1)^{n+1} \frac{\pi}{6}$ and $y = 2n\pi \pm \frac{\pi}{3}$
(iii) $x = n\pi + (-1)^n \frac{\pi}{6}$ and $y = 2n\pi \pm \frac{2\pi}{3}$
(iv) $x = n\pi + (-1)^{n+1} \frac{\pi}{6}$ and $y = 2n\pi \pm \frac{2\pi}{3}$

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (iii) and (iv)

(D) none of these

Q.21

Number of real roots of the equation $\sec\theta + \operatorname{cosec}\theta = \sqrt{15}$ lying between 0 and π is

Q.22

The number of integral values of a for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses a real solution is

Q.23

The number of solution(s) of the equation $(\sin 2x + \cos 2x)^{(1+\sin 4x)} = 2$ in $[-\pi, \pi]$ is

Q.24

The number of solutions of the equation, $\sin^5 x - \cos^5 x = \frac{1}{\cos x} - \frac{1}{\sin x} \quad (\sin x \neq \cos x)$ is

Q.25

If $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$, then $\cos\theta_1 + \cos\theta_2 + \cos\theta_3 =$

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Q.26

If $x^2 - 4x + 5 - \sin y = 0, y \in [0, 2\pi)$, then $\lfloor xy \rfloor =$ (i.e. GIF of xy)

Q.27

The number of solutions of the equation $\cos x + \sin x = 2$ is

Q.28

Number of solution to the equation $\left(\frac{2-\cos^2 x}{\sin x}\right)^3 + \left(\frac{3-\cos 2x}{\sin x}\right) = 0$

Q.29

Let a, b, c be three non-zero real numbers such that equation $\sqrt{3}a\cos x + 2b\sin x = c, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, has distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then, value of $\frac{2b}{a}$ is

Q.30

If $\log_{0.5} \sin x = 1 - \log_{0.5} \cos x$, then number of values of $x \in [-2\pi, 2\pi]$ is

Answers & Solutions

Q.1 Answer:

$\left(0, \frac{\pi}{4}\right)$

Solution:

$\sin x \cdot \cos x \left(\cos^2 x - \sin^2 x\right) > 0$ **or** $\quad \frac{1}{2}\sin 2x \cdot \cos 2x > 0$

or $\quad \sin 4x > 0$

$\therefore \quad 0 < 4x < \pi$

Q.2 Answer:

-1

Solution:

Clearly $-1 \le a \le 1$. For any value of a other than $1, -1$ we know $\sin\theta$ has two values (in quadrants, I, II or III, IV).

Q.3 Answer:

$\frac{\pi}{12}, \frac{11\pi}{36}$

Solution:

$\frac{\sin \pi/12}{\sin x} + \frac{\cos \pi/12}{\cos x} = 2 \quad \Rightarrow \quad \sin\frac{\pi}{12}\cos x + \cos\frac{\pi}{12}\sin x = 2\sin x \cos x$

$\sin\left(\frac{\pi}{12} + x\right) = \sin 2x$

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$\frac{\pi}{12}+x=2x$ **and** $\frac{\pi}{12}+x=\pi-2x$

$\Rightarrow \quad x=\frac{\pi}{12}$ **and** $x=\frac{11\pi}{36}$

Q.4 Answer:

4

Solution:

$\tan x\tan 4x=1\Rightarrow \cos 4x\cos x-\sin 4x\sin x=0$

$\tan x\tan 4x=1$

$\Rightarrow \cos 4x\cos x-\sin 4x\sin x=0$

$\Rightarrow \cos 5x=0$

$\Rightarrow 5x=(2n\pi+)\frac{\pi}{2}n\in Z$

$\Rightarrow x=\frac{(2n+1)\pi}{10};0<x<\pi$

$=\frac{\pi}{10},\frac{3\pi}{10}\frac{\pi}{2}\frac{7\pi}{10}\frac{9\pi}{10}$

But for $\frac{\pi}{2}$, tan x is not defined

Thus, there are 4 solutions only.

Q.5 Answer:

$2n\pi+\frac{7\pi}{4}$

Solution:

$\left.\begin{array}{l}\tan\theta=-1\Rightarrow\theta=\frac{3\pi}{4},\frac{7\pi}{4}\\ \cos\theta=\frac{1}{\sqrt{2}}\Rightarrow\theta=\frac{\pi}{4},\frac{7\pi}{4}\end{array}\right\}\quad\theta=\frac{7\pi}{4}$ **G.s.** $=2n\pi+\frac{7\pi}{4}$

Q.6 Answer:

none of these

Solution:

Let $e^{\cos^2 x}=t\quad\Rightarrow\quad t^2+4t-1=0$

or $\quad t=-2\pm\sqrt{5}$

but $e^{\cos^2 x}>0$ **so** $e^{\cos^2 x}=\sqrt{5}-2$

or $\cos^2 x=\log(\sqrt{5}-2)<0$**. so no solution.**

Q.7 Answer:

$\frac{\pi}{4}$

Solution:

Given, $2\sec 2\alpha=\tan\beta+\cot\beta=\frac{2}{\sin 2\beta}\Rightarrow\cos 2\alpha=\sin 2\beta=\cos\left(\frac{\pi}{2}-2\beta\right)$

$\therefore\quad$ **one solution is** $2\alpha=\frac{\pi}{2}-2\beta$

$\alpha+\beta=\frac{\pi}{4}$

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Q.8 Answer:

$\theta = n\pi; n \in I$

Solution:

$$\sin^2\theta \sec\theta + \sqrt{3}\tan\theta = 0$$
$$\Rightarrow \left(\sin^2\theta + \sqrt{3}\sin\theta\right)\sec\theta = 0$$
$$\Rightarrow \sin\theta(\sin\theta + \sqrt{3})\sec\theta = 0$$
$$\Rightarrow \sin\theta = 0$$
$$\Rightarrow \theta = n\pi, n \in I$$
$$\Rightarrow$$

Q.9 Answer:

$x = n\pi \pm \frac{\pi}{2}; x = m\pi \pm \frac{\pi}{4}; x = k\pi \pm \frac{\pi}{6}$; where $n, m, k \in Z$

Solution:

$$\cos^2 x + \left(2\cos^2 x - 1\right)^2 + \left(4\cos^3 x - 3\cos x\right)^2 = 1$$
$$\Rightarrow \quad 16\cos^6 x - 20\cos^4 x + 6\cos^2 x = 0 \Rightarrow 2\cos^2 x\left(2\cos^2 x - 1\right)\left(4\cos^2 x - 3\right) = 0$$
$$\Rightarrow \quad \cos^2 x = 0; \cos^2 x = \frac{1}{2} = \cos^2\frac{\pi}{4}; \cos^2 x = \frac{3}{4} = \cos^2\frac{\pi}{6}$$
$$\Rightarrow \quad x = n\pi \pm \frac{\pi}{2}; x = m\pi \pm \frac{\pi}{4}; x = kn \pm \frac{\pi}{6}, n, m, k \in Z$$

Q.10 Answer:

$(\cos\theta)^{\sin\theta} < (\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta}$

Solution:

For $\theta \in (\pi/4, \pi/2), 0 < \cos\theta < 1/\sqrt{2} <$ **sin** $\theta < 1$

$\Rightarrow \quad (\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta}$ **and** $(\cos\theta)^{\cos\theta} > (\cos\theta)^{\sin\theta}$ **(as** $0 < \cos\theta < 1$ **and** $\cos\theta < \sin\theta$**)**

Showing that (b) is correct.

Q.11 Answer:

5

Solution:

Given $\left(\frac{1}{2}\right)(\sin 8x + \sin 2x) = \left(\frac{1}{2}\right)(\sin 8x + \sin 4x)$

$\Rightarrow \quad \sin 2x - \sin 4x = 0 \Rightarrow -2\sin x\cos 3x = 0 \Rightarrow \sin x = 0$ **or** $\cos 3x = 0$

That is, $x = n\pi(n \in I)$**, or** $3x = k\pi + \frac{\pi}{2}(k \in I)$.

Therefore, since $x \in [0, \pi]$**, the given equation is satisfied if** $x = 0, \pi, \frac{\pi}{6}, \frac{\pi}{2}$ **or** $\frac{5\pi}{6}$

Q.12 Answer:

$\frac{n\pi}{3} + \frac{\pi}{12}$

Solution:

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$\tan 3x = 1 \Rightarrow \tan 3x = \tan \frac{\pi}{4} \Rightarrow 3x = n\pi + \frac{\pi}{4} \Rightarrow x = \frac{n\pi}{3} + \frac{\pi}{12}$

Q.13 Answer:

$0 < A \leq 1$

Solution:

$0 < \sin^8 \theta \leq \sin^2 \theta$ **...(i)**
and $0 < \cos^{14} \theta \leq \cos^2 \theta$ **...(ii)**
Adding (i) and (ii) $\Rightarrow 0 < A \leq 1$

Q.14 Answer:

$n\pi + \frac{3\pi}{4}, k\pi + \tan^{-1}\left(\frac{4}{3}\right) (n, k \in I)$

Solution:

The given equation is equivalent to
$7\cos^2 x + \sin x \cos x - 3\left(\sin^2 x + \cos^2 x\right) = 0$
or $3\sin^2 x - \sin x \cos x - 4\cos^2 x = 0$
Since $\cos x = 0$ **does not satisfy the given equation we divide throughout by** $\cos^2 x$ **and get**
$3\tan^2 x - \tan x - 4 = 0$
$\Rightarrow \quad (3\tan x - 4)(\tan x + 1) = 0$
$\Rightarrow \tan x = \frac{4}{3}$ or $\tan x = -1$
$x = k\pi + \tan^{-1}\left(\frac{4}{3}\right) \quad x = n\pi + \frac{3\pi}{4} (n, k \in I)$
It should be noted that the solutions given by (b) or (c) are not complete; the general solution is given by (d) only.

Q.15 Answer:

$\frac{9}{4}$

Solution:

$\text{LHS} = \frac{1-\cos 2x}{2} + \frac{1-\cos 2y}{2} + 1 - \cos^2(x+y)$
$= 2 - \frac{1}{2}[\cos 2x + \cos 2y] - \cos^2(x+y)$
$= 2 - \cos(x+y)\cos(x-y) - \cos^2(x+y)$
$= 2 + \frac{1}{4}\cos^2(x-y) - \left[\frac{1}{2}\cos(x-y) + \cos(x+y)\right]^2$
$\leq 2 + \frac{1}{4}(1) - 0$
LHS $\leq \frac{9}{4}$ **(equality occurs if** $x = y = \pi/3$**)**

Q.16 Answer:

none of these

Solution:

$(f(1) < 0 \Rightarrow 3 - 3\sin\theta - 2\cos^2\theta < 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 < 0$
$\Rightarrow \quad \frac{1}{2} < \sin\theta < 1 \Rightarrow 2n\pi + \frac{\pi}{6} < \theta < 2n\pi + \frac{5\pi}{6}$ **and**
$\theta \neq 2n\pi + \frac{\pi}{2}, n \in I)$

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Q.17 Answer:

$[0, \pi]$

Solution:

For real $x, D \geq 0$

$\therefore \quad \cos^2 a - 4(\cos a - 1)\sin a \geq 0$

$\Rightarrow \quad \cos^2 a - 4\sin a \cos a + 4\sin a \geq 0 \Rightarrow$

$(\cos a - 2\sin a)^2 + 4\sin a - 4\sin^2 a \geq 0$

$\Rightarrow \quad (\cos a - 2\sin a)^2 + 4\sin a(1 - \sin a) \geq 0$

$(\cos a - 2\sin a)^2 \geq 0, \text{ and } (1 - \sin a) \geq 0$

$\therefore$ **Inequality (i) will be definitely true when** $0 \leq a \leq \pi$

Q.18 Answer:

$\frac{\pi}{3}$ or $\frac{2\pi}{3}$

Solution:

$1 + \sin\theta + \sin^2\theta + \ldots\ldots\ldots\infty = 4 + 2\sqrt{3}$

$\Rightarrow \quad \frac{1}{(1-\sin\theta)} = 4 + 2\sqrt{3}$

$(1 - \sin\theta) = \frac{(4-2\sqrt{3})}{(4+2\sqrt{3})(4-2\sqrt{3})}$

$= \frac{(4-2\sqrt{3})}{(16-12)} = \frac{(4-2\sqrt{3})}{4}$

$1 - \sin\theta = 1 - \frac{\sqrt{3}}{2}$

$\sin\theta = \frac{\sqrt{3}}{2}$

$\theta = \pi/3, 2\pi/3$

Q.19 Answer:

$x \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

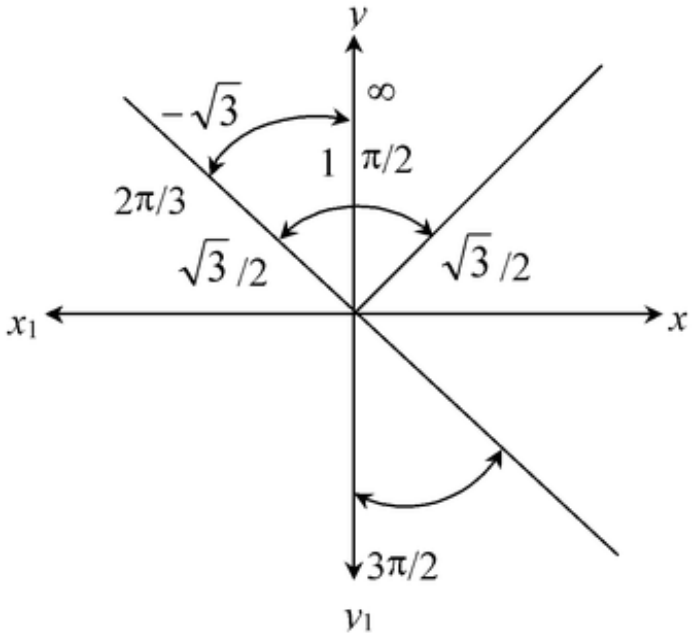
Solution:

$\tan x < -\sqrt{3} \quad \ldots$ **(i) and** $\sin x > \frac{\sqrt{3}}{2}$

(i) $\Rightarrow \frac{\pi}{2} < x < \frac{2\pi}{3}$

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(ii) $\Rightarrow \quad \frac{\pi}{3} < x < \frac{2\pi}{3} \Rightarrow \frac{\pi}{2} < x < \frac{2\pi}{3}$



Q.20 Answer:

(ii) and (iii)

Solution:

We have $2^{\sin x+\cos y} = 1 = 2^0$

$\Rightarrow \quad \sin x + \cos y = 0$ **...(i)**

It is given that

$16^{\sin^2 x+\cos^2 y} = 4 = 16^{1/2}$

$\Rightarrow \quad \sin^2 x + \cos^2 y = \frac{1}{2} \quad ...(ii)$

Eliminating cosy from (i) and (ii), we get

$2\sin^2 x = \frac{1}{2} \Rightarrow \quad \sin x = \pm\frac{1}{2}$

Now, $\sin x = \frac{1}{2} \Rightarrow \quad \cos y = -\frac{1}{2}$

$\Rightarrow \quad x = n\pi + (-1)^n \frac{\pi}{6}$ **and** $\quad y = 2n\pi \pm \frac{2\pi}{3}$

and, $\quad \sin x = -\frac{1}{2} \Rightarrow \cos y = \frac{1}{2}$

$\Rightarrow \quad x = n\pi + (-1)^{n+1} \frac{\pi}{6}$ **and** $\quad y = 2n\pi \pm \frac{\pi}{3}$

Q.21 Answer:

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Solution:

$\sec\theta + \operatorname{cosec}\theta = \sqrt{15} \Rightarrow \sin\theta + \cos\theta = \sqrt{15}\sin\theta\cos\theta \Rightarrow 1 + \sin 2\theta = 15\sin^2\theta\cos^2\theta$

$\Rightarrow \quad 1 + \sin 2\theta = \frac{15}{4}\sin^2 2\theta \quad \Rightarrow 15\sin^2 2\theta - 4\sin 2\theta - 4 = 0 \Rightarrow \quad \sin 2\theta = \frac{2}{3}, \frac{-2}{5}$

Since $0 < \theta < \pi \quad \therefore 0 < 2\theta < 2\pi$

$\sin 2\theta = \frac{2}{5}$ **will give two values of** 2θ **between 0 and** 2π

Hence 4 values of θ **between 0 and** π **are possible.**

Q.22 Answer:

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Solution:

The given equation can be written as

$1-2\sin^2 x+a\sin x=2a-7$

$\Rightarrow \quad 2\sin^2 x-a\sin x+2a-8=0$

$\Rightarrow \sin x=\frac{a\pm\sqrt{a^2-8(2a-8)}}{4}=\frac{a\pm(a-8)}{4}$

$\Rightarrow \sin x=\frac{a-4}{2}$ which is possible if $-1\le\frac{a-4}{2}\le 1$

or $2\le a\le 6$

so the required values of a are $2,3,4,5,6$ and hence the required number is 5.

Q.23 Answer:

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Solution:

L.H.S. ≤ 2, R.H.S. $=2$

Solution is possible only when L.H.S. = R.H.S. = 2

Now L.H.S. $=2$ $\quad\Rightarrow \sin 2x+\cos 2x=\sqrt{2}$ **...(i)**

and $1+\sin 4x=2$ **...(ii)**

From equation (i), $\cos\left(2x-\frac{\pi}{4}\right)=1 \quad\Rightarrow\quad 2x=2n\pi+\frac{\pi}{4}\Rightarrow \quad x=n\pi+\frac{\pi}{8}$

$\Rightarrow x=\frac{\pi}{8},\frac{-7\pi}{8}$. Which satisfies equation (ii).

Q.24 Answer:

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Solution:

The given equation can be written as

$$\sin^5 x-\cos^5 x=\frac{\sin x-\cos x}{\sin x\cos x}$$

$$\sin x\cos x\left[\frac{\sin^5 x-\cos^5 x}{\sin x-\cos x}\right]=1$$

$$\Rightarrow\frac{1}{2}\sin 2x\left[\sin^4 x+\sin^3 x\cos x+\sin^2 x\cos^2 x+\sin x\cos^3 x+\cos^4 x\right]=1$$

$$\Rightarrow \sin 2x\left[\left(\sin^2 x+\cos^2 x\right)^2-2\sin^2 x\cos^2 x+\sin x\cos x\left(\sin^2 x+\cos^2 x\right)+\sin^2 x\cos^2 x\right]$$

$$=2$$

$$\Rightarrow \sin 2x\left[1-\sin^2 x\cos^2 x+\sin x\cos x\right]=2$$

$$\Rightarrow \sin 2x\left[1-\frac{1}{4}\sin^2 2x+\frac{1}{2}\sin 2x\right]=2$$

$$\Rightarrow \sin^3 2x-2\sin^2 2x-4\sin 2x+8=0$$

$$\Rightarrow(\sin 2x-2)^2(\sin 2x+2)=0$$

$\Rightarrow \quad \sin 2x=\pm 2$, **which is not possible for any x.**

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Q.25 Answer:

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Solution:

$\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$

$\Rightarrow \quad \sin\theta_1 = \sin\theta_2 = \sin\theta_3 = 1 \qquad [\text{Q} -1 \leq \sin x \leq 1]$

$\Rightarrow \quad \theta_1 = \theta_2 = \theta_3 = \frac{\pi}{2} \Rightarrow \cos\theta_1 + \cos\theta_2 + \cos\theta_3 = 0$

Q.26 Answer:

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Solution:

$(x-2)^2 + 1 = \sin y$

Since, $1 > \sin y > -1$**, the only possible value of** x **is 2** $\therefore \sin y = 1$

$y = \frac{\pi}{2}$ at $x = 2$

Q.27 Answer:

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Solution:

$|\cos x + \sin x| \leq \sqrt{2}$

and R.H.S. $= 2$**, so no solution**

Q.28 Answer:

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Solution:

This is possible only when $\sin x = \cos x = 1$**, which does not hold simultaneously. Hence, there is no solution.**

Q.29 Answer:

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Solution:

$\Rightarrow \sqrt{3}\cos\alpha + \frac{2\,\text{b}}{\text{a}}\sin\alpha = \frac{\text{c}}{\text{a}} \quad \ldots\ldots(1)$

$\Rightarrow \sqrt{3}\cos\beta + \frac{2\,\text{b}}{\text{a}}\sin\beta = \frac{\text{c}}{\text{a}} \ldots\ldots(2)$

Consider $(1) - (2)$

$\Rightarrow \sqrt{3}[\cos\alpha - \cos\beta] + \frac{2\,\text{b}}{\text{a}}(\sin\alpha - \sin\beta) = 0$

$\Rightarrow \sqrt{3}\left[-2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)\right] +$

$\frac{2\,\text{b}}{\text{a}}\left[2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)\right] = 0$

$\Rightarrow -\sqrt{3} + 2\sqrt{3}\cdot\frac{\text{b}}{\text{a}} = 0$

$\therefore \frac{\text{b}}{\text{a}} = \frac{1}{2} = 0.5.$

$\therefore \frac{2\,\text{b}}{\text{a}} = 1$

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Q.30 Answer:

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Solution:

$\log_{0.5} = 1 - \log_{0.5} \cos x$
$\Rightarrow \log_{0.5}(\sin x) + \log_{05}(\cos x) = 1$
$\Rightarrow \log_{(0.5)}(\sin x \cos x) = 1$
$\Rightarrow \left(\frac{1}{2}\right)^1 = \sin x \cos x$
$\Rightarrow \frac{1}{2} = \frac{1}{2} \times 2 \sin x \cos x$
$\Rightarrow \sin 2x = 1$
$\Rightarrow 2x \in (2x+1)\frac{\pi}{2}$
$\Rightarrow x \in \left[(2n+1)\frac{\pi}{4}\right]$

No.we chat the given ray

$-2\pi \leq x \leq 2\pi$
$\therefore x \in \left[\frac{-7\pi}{4}, \frac{-5\pi}{4}, \frac{-3\pi}{4}, \frac{-\pi}{4}\right]$
$\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

Also, $\log_{0.5}(\sin \mathrm{x}) > 0$
$\log_{0.5}(\cos x) > 0$
$\therefore xt \left[\frac{-7\pi}{4}, \frac{\pi}{4}\right]$

$\therefore$ there are 2 solution in the given rays

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